Automation

DATA201/422 Data Wrangling

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Configuration files

What

`config.yaml`:

```yaml
sex: Female
age: 40-44
n_max: 140
n_draw: 2000
col_fill: lightblue
col_line: darkblue
col_point red
```

How is it used?

```
config <- read_yaml("config.yaml")

df <- data |>
  filter(sex == config$sex,
         age == config$age)
```

Why?

- Single place to look
  - Easier to find
  - Less likely to use conflicting values
- Easier to conduct experiments
- Self-documenting
  - Draw attention to key parameters controlling analysis

Handling errors

Fail fast and loudly

- Fail fast
  - Test inputs as soon as you get them
  - Test calculations as soon as you have made them
  - Easier to diagnose bugs close to their source
- Fail loudly
  - Create an error
    - `stop("Input is wrong")`
    - `raise ValueError("This is an error message.")`
  - Better for a process to fail than to give the wrong answer
- Errors are your friend!

Handling errors

- Sometimes you should modify your pipeline so that it can deal with cases that trigger errors
- Example
  1. Incoming data contains `NULL`s or `NA`s, which trigger errors
  2. Modify your code so that it can deal with `NULL`s or `NA`s (eg by imputing values)
- Make sure you keep testing for cases you aren't covering
  - E.g. keep testing for `NULL`s or `NA`s in variables where you are not imputing
- You might still want to raise a warning
  - `warning("Data has some NAs")`
  - `warnings.warn("Data has some NAs.")`

Continuous integration (CI)

What

- Tests that run when you commit or push
- In GitHub, set up using ‘actions’
- Use AI to help you set them up

Why

- Very commonly used in software development
- But can be useful for data science pipelines
  - Ensures that changes do not break the pipeline

Coping with barriers to automation

Despite benefits of automation, organizations or colleagues can make it difficult

- Organizations might refuse to provide access to tools, eg GitHub, R, Python, Quarto, Jupyter notebooks
- Colleagues might insist on using tools that are difficult/impossible to automate, eg Excel, Word
- People may be unwilling to trust automatic process
- People may feel threatened by automation
  - People derive status by being the only person who can operate a system

Solutions? (Please suggest more!)

- Automate what you can
  - For instance, automate everything up to the report-writing
  - Choose your battles!
- Quarto can generate Word documents…but colleagues will want to edit these
- Reassure people by generating lots of graphs, reports, logs, checks
- Concrete demonstrations are more convincing than abstract arguments